



# Seonghun Bae

ROBOTICS & FUTURE MOBILITY LEARNER

Nam-gu, Ulsan, Republic of Korea

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"Fail forward, grow upward."

## Summary

This is Seonghun Bae, who wants to become robotics engineer seeking to specialize in autonomous navigation and mobile robot (AMR) control systems. My main interests are in ROS2, SLAM, Navigation, simulation and mobile robot control systems. I always try to be a humble learner, listening to feedback from senior engineers and growing together with my colleagues. To improve my skills every day, I carefully record and share my project journeys on GitHub, always ready to take on new challenges and learn from experience.

## Research Interests

**SLAM / Navigation Simulation** Interest in 2D/3D SLAM, Nav2, and autonomous path planning.  
Building robot environments and testing programs in Gazebo, and MuJoCo.

## Education

### Yeungnam University

B.S. IN ROBOTICS ENGINEERING

- Total GPA : 3.79 / 4.5
- Club : ROS Study Group - Academic Member & Project Lead

Gyeongsan, S.Korea

Mar. 2022 - Feb. 2026

## Projects

### Outdoor Mapping Data Optimization using a Quadruped Robot

PARTICIPANT

- Using Deep Robotics Lite3 quadruped robot to make outdoor environment maps - [Notion]
- Testing mapping algorithms and making data better in outdoor environments.
- Fixing sensor errors and getting clean 2D/3D map data from real outdoor roads.
- Learning how to handle 4-legged robot hardware and balance it on uneven ground.

Yeungnam University

Nov. 2024 - Dec. 2025

### Making a Hand Gesture Recognition Model for Robot Control

PARTICIPANT

- Building an AI model to recognize different hand shapes and gestures.
- Testing the model to send movement commands to robots using hand signs.
- Collecting and making hand image datasets to improve the model accuracy.
- Learning how to use computer vision tools and train simple deep learning models.

Yeungnam University

Mar. 2025 - Jun. 2025

### Development of a Mobile Robot with Person Following System

PARTICIPANT

- Building a mobile robot that can find and follow a target person automatically.
- Using ROS2 framework to connect camera sensors and motor control systems.
- Testing the robot in simulation and real rooms to fix path planning errors - [Video]
- Learning how mobile robots move and how to tune motor speeds for safety.

ROS Study Group

Apr. 2024 - Sep. 2024

## Skills

**DevOps** AWS, Docker, Kubernetes, Rancher, Vagrant, Packer, Terraform, Jenkins, CircleCI

**Back-end** Koa, Express, Django, REST API

**Front-end** Hugo, Redux, React, HTML5, LESS, SASS

**Programming** Node.js, Python, JAVA, OCaml, LaTeX

**Languages** Korean, English, Japanese

# Certificates

- Jun. 2026 **Engineer Information Processing (Written)**, HRDK (Human Resources Dev. Service)
- Mar. 2026 **ABEEK Graduate (Washington Accord)**, ABEEK Korea
- Mar. 2026 **Robotics Professional (Accredited)**, Korea Robotics Society
- Nov. 2025 **TOEIC Speaking AL (170)**, YBM
- Apr. 2022 **Computer Literacy (Grade 2)**, KCCI (Chamber of Commerce)
- Dec. 2018 **Type 1 Normal Driver's License**, Korean National Police Agency

# Extracurricular Activity

## GitHub Study Log Repository

*Github Open Source*

OPEN SOURCE CONTRIBUTOR

*Apr. 2026 - Present*

- Operating a public GitHub repository to document daily learning logs - [GitHub]
- Focusing on studying advanced robotics technologies and recording the codes continuously.
- Making study logs open to everyone to share robotics knowledge with global peers.
- Reviewing and refactoring my own source codes to build better software habits.
- Trying to learn core robot systems every day and growing as a passionate engineer.

## Technical Blog

*Naver Blog*

TECHNICAL BLOGGER

*April. 2026 - Present*

- Writing and managing a technical blog to share new engineering knowledge - [Blog]
- Summarizing lectures, reviewing technical papers, and posting helpful development tips.
- Sharing what I have studied to help other developers and remember information longer.
- Contributing to the tech community and making a better society by sharing free knowledge.
- Improving technical writing skills and learning how to explain complex ideas easily.

## ROS Study Group

*Yeungnam University*

PROJECT LEAD & ACADEMIC ORGANIZER

*Apr. 2024 - Dec. 2025*

- Leading and organizing weekly ROS2 study meetings for robotics students.
- Managing team projects and helping members solve hardware and software problems.
- Sharing study materials and robot source codes on GitHub to help the team grow.
- Studying core robotics technology like SLAM, Nav2, and simulation programs together.
- Learning how to collaborate with other software and hardware engineers effectively.